

3(a)	<i>five from</i> 1 drought-tolerant plants have smaller stomatal aperture and lower, transpiration rate / rate of water uptake ; A ora 2 comparative data quote for, mean stomatal aperture / transpiration rate + <u>mmol m⁻² s⁻¹</u> / mean water uptake + <u>cm³</u> per shoot + time + h ; <i>in drought-tolerant plants</i> 3 smaller (aperture), stomata so less water (vapour) loss ; 4 water vapour <u>diffuses</u> (out) through stomata ; 5 less evaporation from (cell walls of) mesophyll ; 6 so less <u>transpiration pull</u> ; 7 AVP ; suggestion of other adaptations of leaves to reduce water loss e.g. sunken stomata / thicker <u>cuticle</u> / hairs / trichomes / lower <u>stomatal density</u> R closed stomata	5
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Question	Answer	Marks
3(b)	I any ref. to mutation / inhibition of (RNA) polymerase 1 microRNA binds to mRNA ; A forms hydrogen bonds with (bases on) mRNA 2 bases in microRNA are <u>complementary</u> to bases on mRNA ; 3 microRNA makes mRNA too large to leave nuclear pore / mRNA cannot reach ribosomes ; <i>mp4 and mp5 accept alternatives to bind</i> 4 mRNA cannot, bind / AW, to (small sub-unit of) ribosome ; A prevents ribosome moving along mRNA 5 anticodons of tRNA cannot, bind / AW, to (some) codons on mRNA ; 6 no / not all, amino acids are brought to ribosome / AW ; 7 AVP ; e.g. complex of microRNA and mRNA recognised for degrading	3